

SESSION 2 · HANDS-ON LAB

COMPLETE GUIDED TUTORIAL

Finding Cyber Threats with NLP and LLMs

An end-to-end Google Colab laboratory using spaCy, NLTK, stylometry and a verification-led LLM workflow

MISSION: Analyse six synthetic messages, rank their investigative priority, and defend every conclusion with evidence.

Duration	Coding level	Installation
45–50 minutes	No prior Python required	No local installation

Participant: _____ Date: _____

Companion notebook supplied with this manual

Session_2_NLP_LLM_Lab_Colab_FINAL.ipynb

Upload this .ipynb file to Google Colab in Section 4. The document also reproduces the essential code so the activity remains understandable without opening the notebook.

Safety and confidentiality rule

Use only the fictional evidence pack in this activity. Never paste live case evidence, personal data, confidential organisational messages, credentials, or legally privileged material into a public Colab notebook or public LLM. Real investigations require an authorised environment and an approved data-handling process.

1 • ACTIVITY OVERVIEW

What You Will Do

You are part of a cyber-forensics analytics team. Six communications have arrived from different channels. Some require urgent investigation; some are benign controls. You will begin with human observation, then use transparent NLP and an LLM to organise, compare and verify the evidence.

Questions your investigation must answer

- ☐ Which people, organisations, places, dates, amounts, email addresses and URLs can be extracted automatically?
- ☐ Which language cues separate malicious or high-risk messages from benign messages?
- ☐ What does a basic stylometric comparison suggest, and what can it not prove?
- ☐ Can an LLM produce a useful threat summary without inventing or merging facts?
- ☐ What final priority and response action should a human investigator recommend?

Process you will follow

1. Understand	2. Read	3. Set up	4. Extract	5. Score	6. Compare	7. Verify	8. Report
Activity	Evidence pack	Colab + tools	spaCy entities	NLTK cues	Stylometry	LLM claims	Final decision

Timing and deliverables

Time	Stage	Output
0–5 min	Read activity and evidence pack	Initial human risk hypothesis
5–12 min	Set up Colab and verify tools	Working spaCy and NLTK environment
12–22 min	Mission 1: spaCy entity extraction	Entity table + missed/wrong notes
22–32 min	Mission 2: NLTK cue scan	Indicator matrix + M1/M5 comparison
32–40 min	Mission 3: stylometry	Similarity lead + limitations
40–47 min	Mission 4: LLM + verification	Groundedness check
47–50 min	Final report	Risk ranking + 45-second briefing

2 • EVIDENCE PACK

Read the Evidence Before Running Any Model

Do not begin with the tool output. First read all six exhibits and form a provisional hypothesis. The ground truth is intentionally withheld from the participant manual and appears only in the instructor answer key.

Forensic habit

Keep the raw message beside every transformed or extracted version. A model table without its source text cannot be independently checked.

EXHIBIT M1 | Executive-payment email

From: "Anil Mehta, CFO" <a.mehta@rneridian-industries.example>

Reply-To: anil.mehta.cfo.office@gmail.example

Subject: URGENT - Confidential wire transfer

Priya,

I am in meetings with the acquisition partners and cannot take calls. We must complete a confidential payment of Rs 48,00,000 to our new vendor Apex Component Traders before 3 PM today, or we lose the contract penalty clause.

Do not discuss this with anyone in the office. Process it under miscellaneous procurement and confirm on this email only. I am counting on you personally.

Anil

EXHIBIT M2 | Remote-job offer

Congratulations!

You have been selected for a remote data-entry position with Global Career Solutions. Earn Rs 35,000 per week from home.

To reserve your onboarding slot, pay a refundable registration fee of Rs 2,999 today and send your Aadhaar copy to jobs.globalcareer.example@gmail.example.

Limited positions available - act immediately!

EXHIBIT M3 | Bank KYC SMS

SBI KYC ALERT:

Your account will be blocked today. Verify PAN and mobile number immediately at <https://bit.ly/SBI-KYC-verify>.

Jaldi karein - link 30 minutes mein expire hoga.

EXHIBIT M4 | Outbreak social post

BREAKING:

A mysterious fever outbreak has closed every school in Pune. Hospitals are hiding the real numbers.

Share this before the authorities delete it! My cousin's neighbour works at a clinic and confirmed 500 cases.

EXHIBIT M5 | Vendor payment reminder

Subject: Vendor payment reminder - Invoice AC-1048

Dear Priya,

This is a reminder that invoice AC-1048 for Rs 1,84,500 falls due on 25 July 2026. The bank details are unchanged from our signed vendor agreement.

Please process it through the usual two-level approval workflow. Contact our accounts desk using the number already held in your vendor master if any clarification is required.

Regards,
Kavita Rao
Apex Components Pvt Ltd.

EXHIBIT M6 | HR announcement

Subject: Updated hybrid-work schedule

Dear colleagues,

From Monday, 27 July 2026, the Pune office will follow the revised hybrid schedule circulated on the HR portal. Team managers will confirm roster days by Friday.

No action is required through email, and HR will never request passwords, payments, or identity documents in response to this notice.

Regards,
Human Resources
Meridian Industries.

Human-first triage (before Colab)

Message	Initial risk: High / Medium / Low	Two clues from the raw text	What would you verify next?
M1	_____	_____	_____
M2	_____	_____	_____
M3	_____	_____	_____
M4	_____	_____	_____
M5	_____	_____	_____
M6	_____	_____	_____

Which message would you investigate first, and why?

3 · YOUR INVESTIGATION BENCH

Tools and the Question Each One Answers

Tool	Role in this lab	The question it helps answer
Google Colab	Runs the notebook in a browser	Can every participant execute the same repeatable analysis?
spaCy	Named Entity Recognition (NER)	Who, where, when, how much, and which organisation?
NLTK	Tokenisation and transparent cue counting	Which visible language cues occur in each message?
Stylometry code	Function-word comparison + simple style statistics	Are two small samples stylistically similar enough to justify a lead?
ChatGPT / approved LLM	Summarisation, extraction and ranking	Can a generative model assist while staying grounded in the exhibits?

Concepts in plain language

Named Entity Recognition (NER)

NER marks spans that look like a person, organisation, place, date, amount or other entity. It accelerates review but can miss Indian names, rupee formats, code-mixed text and defanged URLs.

Lexicon-based cue scanning

A lexicon is a controlled set of words grouped by meaning: urgency, secrecy, finance, authority, threat and reward. The notebook counts visible matches. Counts are explainable, but context still decides meaning.

Stylometry

Stylometry compares measurable writing habits such as function-word rates, sentence length, punctuation and capitalisation. Small samples provide a weak lead, not identification.

Large Language Model (LLM)

An LLM can summarise and organise evidence but may hallucinate, merge details or overstate conclusions. Every factual claim must be checked against M1–M6.

Verification-led workflow

A fluent answer is not automatically a reliable answer. A human investigator labels every claim supported, partly supported, unsupported or missed and owns the final report.

4 • COLAB SETUP

Google Colab: From Zero to a Working Lab

Open [Google Colab](#) in Chrome, Edge or Firefox and sign in with a Google account.

Step 1. Open the upload dialog — Choose File → Upload notebook.

Step 2. Select the companion file — Choose Session_2_NLP_LLM_Lab_Colab_FINAL.ipynb from your computer.

Step 3. Save your own copy — Choose File → Save a copy in Drive. Work in the copy so outputs are retained.

Step 4. Connect to a runtime — Click Connect in the upper-right corner if Colab has not connected automatically.

Step 5. Run the installation cell — Click the ▶ button beside Step 0. Wait until Setup complete appears.

Step 6. Run the verification cell — Click the next ▶ button. Confirm version numbers, spaCy entities and an NLTK token list.

Step 7. Load the evidence pack — Run the messages cell. It should print: Loaded 6 synthetic messages.

Step 8. Proceed mission by mission — Run only the cell for the current mission, record the output, and answer the questions before moving on.

What the Colab screen means

Control / indicator	Meaning	Use in this lab
▶ beside a code cell	Execute this cell	Run one mission at a time
Spinning indicator	Cell is still running	Wait; do not click repeatedly
Green check	Cell completed successfully	Read the output below
Red traceback	Cell failed	Read the last line and use troubleshooting
Runtime → Run all	Run every cell from top to bottom	Use only after setup succeeds
Runtime → Restart session	Clear memory and start again	Use after package/state problems
Files icon	Shows generated CSV files	Download results at the end

Installation cell: what it does

COLAB CELL 0 — INSTALL

```
%pip -q install spacy nltk pandas scikit-learn
!python -m spacy download en_core_web_sm -q
print("Setup complete.")
```

The first line installs the Python packages inside the temporary Colab runtime. The second downloads spaCy's small English language model. The runtime is temporary; repeat this cell when a new session begins.

Successful setup checklist

- ☐ The install cell ends with "Setup complete."
- ☐ The verification cell prints version numbers.
- ☐ The spaCy demonstration prints one or more entities.

- ☐ The NLTK demonstration prints a list of tokens.
- ☐ The evidence cell confirms that six messages are loaded.

Do not use “Run all” too early

Run cells from the top in order. Use Runtime → Run all only after the setup and evidence cells succeed.

5 • TOOL TUTORIAL

How spaCy Is Installed, Loaded and Called

Installing, importing, loading and calling are different actions. The notebook shows all four explicitly.

Action	Exact code	Meaning
Install	%pip install spacy	Make the package available in this Colab runtime.
Download model	!python -m spacy download en_core_web_sm	Obtain a trained English language pipeline.
Import	import spacy	Give the notebook access to the package.
Load	nlp = spacy.load("en_core_web_sm")	Create the trained pipeline object.
Call	doc = nlp(text)	Send one message through the pipeline.
Read output	for ent in doc.ents: ...	Inspect the entities returned by the model.

SPACY DEMONSTRATION CELL

```
import spacy

nlp = spacy.load("en_core_web_sm")
text = "Anil Mehta requested Rs 48,00,000 in Pune on 20 July 2026."

doc = nlp(text)          # CALL spaCy

for ent in doc.ents:
    print(ent.text, ent.label_)

`nlp(text)` is the actual model call. `doc.ents` contains the predicted entity spans.
```

Entity labels used in this activity

Label	Meaning	Possible lab example
PERSON	Named person	Anil Mehta; Kavita Rao
ORG	Organisation or institution	Meridian Industries; SBI
GPE	Country, city or state	Pune
DATE / TIME	Calendar or time expression	25 July 2026; 3 PM today
MONEY	Monetary value	Rs 48,00,000
CARDINAL	Number not assigned another label	500
ARTEFACT	Custom email/URL-like extraction added by	hxxps://bit[.]ly/...

Label	Meaning	Possible lab example
	the notebook	
Forensic caution spaCy predicts; it does not “know” the case. A missed name, wrongly labelled amount or unrecognised defanged URL is a model limitation that must be recorded, not silently corrected in the output table.		

6 • TOOL TUTORIAL

How NLTK Is Imported and Called

The final notebook uses NLTK’s TreebankWordTokenizer. It requires no extra tokenizer-data download, making the lab more reliable. The familiar word_tokenize alternative is shown in the note below.

Action	Exact code	Meaning
Install	%pip install nltk	Make NLTK available in the runtime.
Import tokenizer	from nltk.tokenize import TreebankWordTokenizer	Load a tokenizer class.
Create tokenizer	tokenizer = TreebankWordTokenizer()	Create the tokenizer object.
Call	tokens = tokenizer.tokenize(text)	Split the message into tokens.
Normalise	token.lower()	Allow case-insensitive lexicon matching.
Count	sum(token in word_set ...)	Count visible cue matches.

NLTK DEMONSTRATION CELL

```
from nltk.tokenize import TreebankWordTokenizer
```

```
tokenizer = TreebankWordTokenizer()
```

```
text = "URGENT: complete the confidential payment today!"
```

```
tokens = tokenizer.tokenize(text)    # CALL NLTK
```

```
print(tokens)
```

The output is a Python list of words and punctuation. Tokenisation is the first step before cue matching or stylometry.

Alternative used in many tutorials

You may also use `from nltk.tokenize import word\_tokenize` and then `word\_tokenize(text)`. That version normally requires `nltk.download("punkt")` and, in recent NLTK versions, `nltk.download("punkt\_tab")`.

Cue families used by the notebook

Cue family	Example terms	What it may indicate
Urgency	urgent, immediately, today, blocked, before	Time pressure that reduces verification
Secrecy	confidential, do not discuss, personally	Isolation from colleagues or controls

Cue family	Example terms	What it may indicate
Finance	payment, transfer, fee, invoice, bank, account	Financial context; not malicious by itself
Authority	CFO, HR, bank, authorities	Trust or rank invocation
Threat / pressure	blocked, lose, penalty, delete, hiding	Fear or consequence framing
Reward	congratulations, selected, earn, opportunity	Reward framing and advance-fee patterns

Interpretation rule

One cue is not a verdict. Compare clusters and context. M1 and M5 both contain finance terms; the surrounding secrecy, urgency, process safeguards and verification instructions distinguish them.

7 · MISSION 1 · 10 MINUTES

Extract Digital Evidence with spaCy

Objective: transform M1–M6 into a structured entity table and record model misses or wrong labels honestly.

Exact steps in Colab

1. Scroll to “Mission 1 — Extract Digital Evidence with spaCy”.
2. Run the main code cell. Wait for a pandas table named entities_df.
3. Read the Message, Entity, Label and Meaning columns.
4. Run the optional filter cell for M1, then change M1 to M2, M3 and so on if needed.
5. Compare every model prediction against the original evidence exhibit.
6. Complete the table below. Record both useful hits and important errors.

CORE SPACY CALL INSIDE THE MISSION

```
for message_id, text in messages.items():
    doc = nlp(text)          # CALL spaCy
    for ent in doc.ents:
        print(message_id, ent.text, ent.label_)
```

The full notebook also uses a regular expression to recover email and URL-like artefacts that general NER may miss.

Participant entity record

ID	People / organisations	Date / place	Money / number	Email / URL artefact	Missed or wrong?
M1	_____	_____	_____	_____	_____
M2	_____	_____	_____	_____	_____
M3	_____	_____	_____	_____	_____
M4	_____	_____	_____	_____	_____
M5	_____	_____	_____	_____	_____

ID	People / organisations	Date / place	Money / number	Email / URL artefact	Missed or wrong?
M6	_____	_____	_____	_____	_____

Interpretation questions

Did spaCy recognise Anil Mehta and Apex Component Traders correctly?

Did it recognise Rs 48,00,000 and Rs 1,84,500 as MONEY?

How did it handle the look-alike email domain and the defanged URL?

Which missed entity would matter most to an investigator?

Why must the raw message remain attached to the entity table?

Detect Suspicious Language with NLTK

Objective: use an explainable lexicon to count cue families across all six messages, then interpret the counts in context.

Exact steps in Colab

1. Scroll to “Mission 2 — Score Suspicious Language with NLTK”.
2. Run the scanner cell. A table named `indicator_df` appears.
3. Record urgency, secrecy, finance, authority, threat_pressure, reward, caps_ratio, exclamations and total_cue_hits.
4. Compare M1 with M5. Both contain payment language; inspect the other columns.
5. Run the extension cell that adds ``jaldi`` and ``karein`` to the urgency lexicon.
6. Record how M3 changes and explain why local-language coverage matters.

CORE NLTK CALL AND CUE COUNT

```
raw_tokens = tokenizer.tokenize(text)    # CALL NLTK

tokens = [t.lower() for t in raw_tokens
          if any(ch.isalnum() for ch in t)]

score = {
    family: sum(token in words for token in tokens)
    for family, words in LEX.items()
}
```

Participant cue matrix

ID	Urgency	Secrecy	Finance	Authority	Threat	Reward	Human interpretation
M1	—	—	—	—	—	—	_____
M2	—	—	—	—	—	—	_____
M3	—	—	—	—	—	—	_____
M4	—	—	—	—	—	—	_____
M5	—	—	—	—	—	—	_____
M6	—	—	—	—	—	—	_____

Required comparison

Why is M1 more suspicious than M5 even though both contain financial vocabulary?

What false positive could a finance-only keyword filter create?

What changed after adding the Hinglish terms, and why?

9 · MISSION 3 · 8 MINUTES

Compare Writing Style: A First Stylometric Lead

Objective: compare function-word profiles using cosine similarity and inspect sentence length, exclamation use and capitalisation separately.

Important limitation

These messages are very short and belong to different genres. The output is a classroom demonstration of comparative features, not a valid identification of an author.

Exact steps in Colab

- 1. Run the Mission 3 cell and locate stylometry_df.
- 2. Record the function-word cosine similarity for each supplied pair.
- 3. Read the separate style statistics for each message.
- 4. Identify the highest numerical similarity.
- 5. Write a calibrated conclusion and at least two alternative explanations.

FUNCTION-WORD COSINE SIMILARITY

```
profile_a = function_word_profile(messages["M1"])
profile_b = function_word_profile(messages["M5"])

similarity = cosine_similarity([profile_a], [profile_b])[0, 0]
print(round(float(similarity), 3))
```

A higher value means more similar profiles under this narrow feature set; it does not establish common authorship.

Pair	Similarity	Sentence / punctuation observation	Calibrated interpretation	Alternative explanation
M1 vs M2	_____	_____	_____	_____
M1 vs M5	_____	_____	_____	_____
M5 vs M6	_____	_____	_____	_____
M2 vs M3	_____	_____	_____	_____

Use courtroom-safe language: "The texts are stylistically ..."

What additional known samples and controls would a real analysis require?

Use an LLM, Then Verify It

Objective: test whether an LLM can assist an investigator while remaining grounded in the evidence pack.

Exact steps

1. Open ChatGPT or another institutionally approved LLM in a separate browser tab.
2. Start a new conversation.
3. Copy M1–M6 from this manual or the notebook. Use only this fictional evidence.
4. Run the notebook cell that prints LLM_PROMPT, then copy the complete prompt.
5. Paste the evidence and prompt together and submit.
6. Check every factual claim against the cited exhibit before accepting it.
7. Count supported, partly supported, unsupported and missed items.

PROMPT TO USE

You are assisting a cyber investigation. Analyse M1–M6.

1. Extract people, organisations, locations, dates, amounts, email addresses and URLs.
 2. State each message's apparent purpose.
 3. Identify urgency, secrecy, authority, reward, threat, financial requests, safeguards or amplification tactics.
 4. Rank M1–M6 from highest to lowest investigative priority.
 5. Cite M1–M6 for every factual claim.
 6. If a fact is not directly stated, write "not stated".
- Do not infer identity, motive, linkage or truthfulness.

Return an entity table, risk-ranking table, analyst summary, and limitations / uncertainties.

Verification table

LLM claim	Message cited	Supported / Partial / Unsupported	Exact exhibit text	Correction or missed item
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

Groundedness scorecard

Supported	Partly supported	Unsupported	Important missed items	Overall judgement
—	—	—	—	Reliable / Needs correction / Unreliable

Did the LLM merge details or invent identity, motive, linkage or technical facts?

Would you sign the LLM report without correction? Why?

Confidentiality reminder

This lab pack is fictional. Real case evidence must not enter a public LLM without explicit authorisation, approved contractual safeguards and an evidence-handling process.

11 • FINAL DELIVERABLE

Final Investigator Report

Combine human reading, NLP output and verified LLM assistance. The final decision belongs to you, not to a model.

ID	Human assessment	Strongest NLP evidence	Verified LLM contribution	Final priority	Recommended action
M1	—	—	—	—	—
M2	—	—	—	—	—
M3	—	—	—	—	—
M4	—	—	—	—	—
M5	—	—	—	—	—
M6	—	—	—	—	—

Final ranking

1. Message ___ — reason:

2. Message ___ — reason:

3. Message ___ — reason:

4. Message ___ — reason:

5. Message ___ — reason:

6. Message ____ — reason:

45-second investigator briefing

Use this structure: (1) your overall claim, (2) the three strongest facts, (3) the main AI limitation, and (4) the recommended action.

Briefing notes:

Recommended action options

- ☐ Escalate immediately for technical and organisational investigation.
- ☐ Quarantine or block pending validation.
- ☐ Request independent verification using a trusted channel.
- ☐ Preserve and monitor; further corroboration required.
- ☐ No immediate action; retain as a benign control.

Troubleshooting

Problem	Likely cause	Exact action
No module named spaCy / NLTK	Install cell was skipped	Run Step 0, then restart and run from the top.
Cannot find en_core_web_sm	Model download failed	Run <code>!python -m spacy download en_core_web_sm -q`</code> .
name nlp is not defined	spaCy load cell was skipped	Run the environment verification cell.
name messages is not defined	Evidence cell was skipped	Run Step 2 before the missions.
Table is empty	An earlier cell failed	Read the last red error, fix, then rerun above cells.
Colab disconnected	Idle timeout or network interruption	Reconnect and use Runtime → Run all.
LLM gives no citations	Prompt was shortened or ignored	Resubmit the exact prompt and require M1–M6 citations.
LLM refuses / warns	Platform policy or institutional restriction	Use an approved model or complete the task manually.

Quick glossary

Term	Meaning in this lab
Token	A word, punctuation item or symbol produced by a tokenizer.

Term	Meaning in this lab
Entity	A span predicted as a person, organisation, place, date, amount, etc.
Lexicon	A controlled word set used for transparent cue matching.
Feature	A measurable property used in analysis.
Cosine similarity	A measure of how similarly two numerical profiles point.
False positive	A benign message incorrectly treated as suspicious.
Hallucination	An LLM statement not supported by the supplied evidence.
Grounded claim	A claim directly supported by a cited exhibit.

Completion checklist

- ☐ I retained the original evidence pack beside every model output.
- ☐ I recorded model misses and wrong labels instead of hiding them.
- ☐ I interpreted cue counts in context and checked benign controls.
- ☐ I described stylometry as a lead, not proof.
- ☐ I verified every LLM claim before using it in the final report.

One-minute reflection

Which tool produced the most useful investigative lead, and why?

Which model failure surprised you most?

What would you change before using this workflow on a real case?
